

127. Word Ladder

Hard

Topics

Companies

A **transformation sequence** from word `beginWord` to word `endWord` using a dictionary `wordList` is a sequence of words `beginWord -> s1 -> s2 -> ... -> sk` such that:

- Every adjacent pair of words differs by a single letter.
- Every `si` for `1 <= i <= k` is in `wordList`. Note that `beginWord` does not need to be in `wordList`.
- `sk == endWord`

Given two words, `beginWord` and `endWord`, and a dictionary `wordList`, return *the number of words in the shortest transformation sequence from `beginWord` to `endWord`, or 0 if no such sequence exists.*

Example 1:

Input: `beginWord = "hit", endWord = "cog", wordList = ["hot","dot","dog","lot","log","cog"]`

Output: 5

Explanation: One shortest transformation sequence is "hit" -> "hot" -> "dot" -> "dog" -> "cog", which is 5 words long.

Example 2:

Input: `beginWord = "hit", endWord = "cog", wordList = ["hot","dot","dog","lot","log"]`

Output: 0

Explanation: The endWord "cog" is not in wordList, therefore there is no valid transformation sequence.

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Explanation in Telugu Roman Script (English Letters)

Ippudu ide concept ni simple ga Telugu lo step-by-step ardam cheskundam.

1. Core Logic (Gurtu pettukovaalsina mukhyamaina vishayam)

Ee problem ni "Words" laaga kakunda oka **Map (Graph)** laaga aalochinchali.

- Prati word oka city anukondi.
- Okka letter matrame change ayye rendu words madhya matrame road untundi. Example: "hit" nunchi "hot" ki vellochu, endukante okka 'i' matrame 'o' ga marindi. But "hit" nunchi "dot" ki direct ga vellelem.
- Mana goal: `beginWord` nunchi `endWord` ki **Shortest Path** (atanni thakkuva steps lo) vellali.
- Graph lo eppudaina "Shortest Path" adigithe, blind ga gurtu pettukovalsina oke okka technique: **BFS (Breadth-First Search)**. BFS enduku use chestham ante, idi water lo ripples (alaalu) laaga chuttu pakkala unna anni dhaarula ni oke saari vethukuthu velthundi. Kabatti target dhorikinappudu adi khachitanga shortest path ye avuthundi.

2. Step-by-Step Em Jaruguthundi?

- Queue and Set:** Modata mana `wordList` ni oka `unordered_set` (dict) lo pettukuntam, dheni valla oka word andulo undho ledho thwaraga check cheyochu.
- Queue lo Start Word:** Mana `beginWord` ni queue lo vesi, count (`level`) ni 1 ga start chestham.
- Looping:** Queue khali ayye varaku loop run chestham.
- Letter Changing Logic:** Queue lo unna prati word ni tiskoni, dhani lo unna prati letter place lo 'a' nunchi 'z' varaku petti check chestham.
 - Example: "hit" unte... "ait", "bit", "cit" ... ala "hot" vachinapudu, adi mana dictionary set lo unte, dhanni queue lo vesi, dictionary nunchi delete chestham (malli repeat avvakunda undataniki).
- Goal Reached:** Epudaina mana queue nunchi bayataku vachina word `endWord` ki equal aithe, ventane `level` return chestham. Ade mana answer.

3. Base Cases & Edge Cases ela handle chestham?

- Target Leka pothe:** Asalu `endWord` anedhi `wordList` lo leka pothe, mana goal reach avvadam impossible. Anduke starting lone check chesi `return 0` chestham.
- Visited Words:** Okka saari visit chesina word ni malli visit chesthe loop infinite ayyothundi. Anduke dictionary `set` nunchi ventane `erase()` (remove) chesi handle chestham.

126. Word Ladder II

Hard

Topics

Companies

A **transformation sequence** from word `beginWord` to word `endWord` using a dictionary `wordList` is a sequence of words `beginWord -> s1 -> s2 -> ... -> sk` such that:

- Every adjacent pair of words differs by a single letter.
- Every `si` for `1 <= i <= k` is in `wordList`. Note that `beginWord` does not need to be in `wordList`.
- `sk == endWord`

Given two words, `beginWord` and `endWord`, and a dictionary `wordList`, return *all the shortest transformation sequences* from `beginWord` to `endWord`, or an empty list if no such sequence exists. Each sequence should be returned as a list of the words `[beginWord, s1, s2, ..., sk]`.

Example 1:

Input: `beginWord = "hit", endWord = "cog", wordList = ["hot","dot","dog","lot","log","cog"]`

Output: `[["hit","hot","dot","dog","cog"],["hit","hot","lot","log","cog"]]`

Explanation: There are 2 shortest transformation sequences:

`"hit" -> "hot" -> "dot" -> "dog" -> "cog"`

`"hit" -> "hot" -> "lot" -> "log" -> "cog"`

Example 2:

Input: `beginWord = "hit", endWord = "cog", wordList = ["hot","dot","dog","lot","log"]`

Output: `[]`

Explanation: The endWord "cog" is not in wordList, therefore there is no valid transformation sequence.

Explanation in Telugu Roman Script (English Letters)

Ee Word Ladder II problem ni Telugu lo chala clear ga ardam cheskundam.

1. Core Logic (Asalu em jaruguthundi?)

Word Ladder I ki Word Ladder II ki unna taddela entante: Word Ladder I lo mana ki kevalam enni steps pattindi (`length`) ane మొత్తం matrame kaavali. Kani Word Ladder II lo enni **valid shortest dhaarulu (paths)** unnayo, vatini annitini list laaga return cheyali.

Idhi cheyadaniki manam rendu techniques mix chestham: **BFS + DFS (Backtracking)**.

- **BFS enduku?** Anni paths ni direct ga vethikithe time and memory aypothundi (TLE/MLE vasthundi). Anduke modata BFS run chesi, start word nunchi prati word ki unna accurate level distance ni oka map (`steps`) lo record chestham.
- **DFS enduku?** Map ready ayyaka, target `endWord` nunchi reverse lo (`endWord -> ... -> beginWord`) venakki travel chestham. Ee travel chesestundi, kevalam `level - 1` unna words lokene veltham. Dheni valla wrong roots loki vellakunda directly short paths matrame track chestham.

2. Step-by-Step Logic

- **BFS & Level Mapping:** `beginWord` ni level 1 anukuntam. Dheni nunchi generate ayye words ni level 2, level 3 laaga queue dwara step count mapping chestham.
- **Backtrack via DFS:** Map tayaruvayi, manam `endWord` daggara start chestham (pina code lo reverse tracing chesam convenient ga undataniki).
- **Valid Neighbor Verification:** Ippudu `"cog"` level 5 aithe, dhani neighbor `"dog"` level 4 lo undho ledho check chestham. `4 == 5 - 1` kabatti adi valid path. Alage step by step level tagguthu `beginWord` reach avvagane aa full path ni `ans` vector lo save chestham.

3. Edge Cases

- **Target Leka pothe:** `endWord` asalu `wordList` lo leka pothe output direct ga empty list `[]` pethali.
- **Same level sharing:** Okate word ki rendu valid parents undochu (Example: `"dog"` and `"log"` rendu `"cog"` ki reach avvachu). BFS run chesetapudu, a dharulani miss avvakunda manam map lo core levels ni manage chestham.

130. Surrounded Regions

Medium

Topics

Companies

You are given an $m \times n$ matrix `board` containing letters `'X'` and `'O'`, capture regions that are **surrounded**:

- **Connect**: A cell is connected to adjacent cells horizontally or vertically.
- **Region**: To form a region **connect every** `'O'` cell.
- **Surround**: A region is surrounded if none of the `'O'` cells in that region are on the edge of the board. Such regions are **completely enclosed** by `'X'` cells.

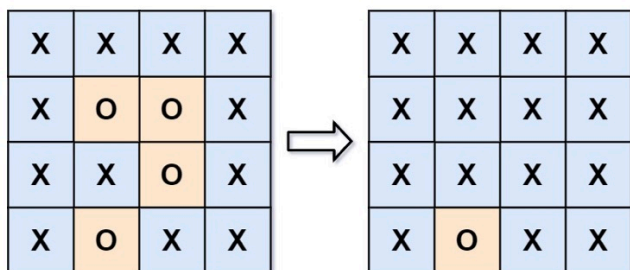
To capture a **surrounded region**, replace all `'O'`s with `'X'`s **in-place** within the original board. You do not need to return anything.

Example 1:

Input: `board = [ ["X","X","X","X"], ["X","O","O","X"], ["X","X","O","X"], ["X","O","X","X"] ]`

Output: `[ ["X","X","X","X"], ["X","X","X","X"], ["X","X","X","X"], ["X","O","X","X"] ]`

Explanation:



In the above diagram, the bottom region is not captured because it is on the edge of the board and cannot be surrounded.

Example 2:

Input: `board = [ ["X"] ]`

Output: `[ ["X"] ]`

Jun 11

Part 2: Explanation in Telugu Roman Literature

1. Core Logic (Gurtu pettukovaalsina mukhyamaina vishayam)

Ee problem ni sulabhamga gurtu pettukovali ante **Reverse Logic (Vennaki aalochinchadam)** vaadali.

- Grid madyalo unna '0' s ni vethukuntu velladam kante, border (anchulu) meeda unna '0' s ni chudadam chala easy.
- Rule entante:** '0' ayina boundary ki touch avvuthunte, dhanni 'X' tho kapotam (surround cheyadam) kudaradu. Adi eppudu "Safe" ga untundi.
- Dani pakkana unna inko '0' kuda automatic ga safe avuthundi.

2. Step-by-Step Em Jaruguthundi?

- Boundary Scan:** Manam modata grid yokka 4 borders (pina, kinda, left, right edges) meeda unna '0' s ni matrame vethukutam.
- DFS Flooding:** Boundary meeda ekkadaina '0' kanipisthe, ventane akkada nunchi DFS start chesi lopali vaipuki veltham. Alaa connectivity unna anni '0' s ni temporary ga oka dummy character '#' ga marchesi "Safe" ani mark chestham.
- Final Transformation:** Ippudu mottam grid ni chustham:
 - Ekadaithe inka '0' migili poyindho, dhaaniki boundary ki ae connection ledhu ani ardham (trapped!). Kabatti vatini 'X' ga marchesi capture chestham.
 - Ekadaithe '#' mark undho, avi safe regions. Kabatti vatini malli mamulga '0' ga marchesi unchutham.

3. Base Cases Ela Handle Chestham?

DFS recursive function run ayye loop avvakunda aagipovanni conditions:

- Mana row coordinate 0 kante thakkuva unna, leda total rows kante ekkuva unna ($r < 0$ || $r \geq \text{rows}$).
- Mana column coordinate boundary cross ayina ($c < 0$ || $c \geq \text{cols}$).
- Mana cell character '0' kakunda, adi kuda 'X' leda already visit ayina '#' character ayina.

Ee conditions trigger avvagane function safe ga return ayyi infinite runs ni aapesthundi. Ee logic chala strong and easy to remember!

+ Ask Gemini

Flash



Gemini is AI and can make mistakes.

Jun 11

